Turgut Aghashov

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Research Interests: Security of LLM-backed agents (prompt injection, agentic AI in CI/CD and developer workflows) · AI for security (LLM-assisted vulnerability discovery and code analysis) · Web and systems security · Kernel instrumentation and program analysis

EDUCATION

University of Maryland College Park, MD, USA
Professional Master of Engineering, Cybersecurity Engineering Aug 2024 - Dec 2025

- **Relevant coursework:** Hacking of C Programs and Unix Binaries, Networks and Protocols, Cloud Computing, Advanced Hacking of Linux, Network Security, Reverse Engineering, Fundamentals of AI and Deep Learning, Intro to Secure Coding, Secure Operating Systems.
- **GPA:** 3.88/4.0

Baku Higher Oil School Baku, Azerbaijan
Bachelor of Engineering with Honors, Information Security Sep 2019 - Jun 2024

- **Thesis:** Mobile Security Ecosystem and Penetration Testing (see Research Experience).
- **Relevant coursework:** Web and Mobile Programming and Security, Database Security, Network Security, Data Structures and Algorithms.
- **GPA:** 95.57/100

PUBLICATIONS

1. **T. Aghashov**. "Depth of Compromise: Measuring Exploitation-Intent Depth of Indirect Prompt Injection Across LLM-Backed CI/CD Agents." Manuscript in preparation, 2026.
2. **T. Aghashov**. "Mobile Security Ecosystem and Penetration Testing." 5th International Scientific Conference of Students and Young Researchers, Baku Higher Oil School, Apr-May 2024, pp. 467-469. ISBN 978-9952-8513-0-4. Link
3. **T. Aghashov** and N. Najafli. "Building an Effective Security Operation Center with Open-Source Tools." 4th International Scientific Conference of Students and Young Researchers, Baku Higher Oil School, Apr-May 2023, pp. 340-341. Link

RESEARCH EXPERIENCE

Depth of Compromise: Indirect Prompt Injection Depth in LLM-Backed CI/CD Agents - GitHub
Independent Research Aug 2026 - present

- Proposed a 5-rung exploitation ladder (canary echo → tool call → secret read → exfiltration → CI-config write) that measures how deep an injection drives an agent's intent, instead of binary success.
- Built an air-gapped testbed of three CI/CD agents (code review, PR summarization, issue triage) with simulated tools and a decision-level scoring oracle.
- Ran 16,327 trials across 11 models from four vendors, using 99 payloads spanning 7 injection vectors and 12 technique families.
- Found that susceptibility is strongly model-specific, with techniques that work on one vendor often failing on another, and that a model's ability to detect the injection, not its reasoning architecture, best predicts resistance.

Internship Research - RefProbe: Hunting Kernel Lifetime Bugs with eBPF - GitHub
Bureau Veritas Cybersecurity North America, mentor: Emmanuel Gambliel Jul 2025 - Aug 2025

- Built an eBPF tracer that detects Linux kernel lifetime bugs (use-after-free, double-free) by correlating allocation and deallocation events to flag dangling-pointer access.
- Engineered test kernel modules to inject controlled bugs and measure detection coverage and false positives.
- Achieved 87% detection on injected lifetime bugs with <5% false positives.

Undergraduate Thesis Research - Mobile Security Ecosystem and Penetration Testing
Baku Higher Oil School Feb 2024 - May 2024

- Assessed an Android banking app across application, OS, and network layers with static and dynamic analysis (MobSF, Frida, Drozer, Burp Suite).
- Mapped findings to the OWASP Mobile Top 10 and demonstrated client/server exploit chains exposing financial data.
- Presented at the 5th International Scientific Conference of Students and Young Researchers (BHOS, 2024).

TEACHING EXPERIENCE

Instructor, Information Security Fundamentals - TechBridge Academy Jul 2026 - Sep 2026

- Taught a 16-session course for students aged 12-17 in two modules: Internet and Networks (CIA triad, TCP/IP and DNS, Wireshark packet analysis, account security) and Linux and Security (shell, permissions and processes, OSINT, malware basics, cryptography and steganography, Bash scripting).
- Ran hands-on labs with Wireshark, OSINT tooling, openssl, steghide, and CyberChef, closing the first module with a packet-hunting mini-CTF and the second module with cryptography and Linux projects.

Instructor, Offensive Security Program - Log Academy Jul 2026 - Aug 2026

- Taught a 9-lesson penetration testing course to 12 students, progressing from Linux foundations and pentest methodology (MITRE ATT&CK, rules of engagement, recon) to Linux and Windows/Active Directory exploitation.
- Covered the full attack chain on each platform: enumeration, initial access, privilege escalation, credential attacks, persistence, and lateral movement.
- Designed a 2-hour hands-on lab and homework for every session, plus a capstone engagement with reporting standards.

Math Coach - Math Success Program, UMD Oct 2025 - Dec 2025

- Tutored undergraduates in precalculus and calculus (MATH 113/120/140/141) in one-on-one and small-group sessions.

Red Team Instructor - Cyber Club, Baku Higher Oil School Jan 2024 - Jun 2024

- Delivered offensive-security training for junior students: reconnaissance, web exploitation, privilege escalation.

Cryptography Teaching Assistant - Azerbaijan Cyber Security Organizations Association Jun 2023 - Jul 2023

- Tutored 30 students in understanding key concepts and practical applications in cryptography.

SELECTED PROJECTS

Man-in-the-Middle Attacks on the Southbound Interface in SDNs - ENPM 693 Network Security, UMD - GitHub

- Built a CORE + Ryu-based SDN testbed to emulate controller-switch communication over OpenFlow.
- Implemented ARP-spoofing and traffic-relay attacks to intercept and modify southbound traffic in real time.
- Evaluated detection and mitigation strategies using flow-rule validation, anomaly monitoring, and controller-side checks.

gdb_ghpseudo: Ghidra Pseudocode Integration for GDB - open-source tool - GitHub

- Shows synchronized Ghidra pseudocode in gdb/pwndbg while stepping through stripped binaries (headless Ghidra server).

Secure AWS Cloud Architecture - ENPM 818N Cloud Computing, UMD - GitHub

- Designed a multi-AZ VPC with an ALB fronting autoscaling web servers and MariaDB, hardened with AWS WAF.

INDUSTRY EXPERIENCE

Leading Penetration Testing Specialist - Innovation and Digital Development Agency, Baku Apr 2026 - present

- Conduct pre-release penetration tests of new features and third-party integrations (web, mobile, API, infrastructure).
- Triage findings with developers and project managers, prioritizing critical/high vulnerabilities and guiding remediation.
- Perform adversarial testing of AI-enabled features (prompt injection, jailbreaks); motivated the Depth of Compromise study.

Security Engineering Intern - Bureau Veritas Cybersecurity, Seattle, WA Jun 2025 - Aug 2025

- Identified 150+ security flaws through advanced web & cloud penetration testing lab environments.
- Assessed 3 enterprise web/infrastructure systems, uncovering critical misconfigurations and producing remediation reports.

Penetration Testing Specialist - Cyberpoint LLC, Baku Dec 2023 - Aug 2024

- Executed black-, gray-, and white-box pentests for 5 banks, 2 enterprises, and 2 telecoms; found 100+ web vulnerabilities.
- Instrumented Android/iOS apps with Frida and Objection to bypass pinning and root detection for dynamic analysis.

Junior Cyber Research and Assessment Specialist - Prosol CJSC, Baku Aug 2022 - Nov 2023

- Conducted web, mobile, and infrastructure penetration tests across 4 banks and 2 enterprises.
- Traced and analyzed 5 phishing campaigns targeting government agencies, recommending defensive measures.

CERTIFICATIONS

OffSec Web Expert (OSWE), Oct 2024 · OffSec Certified Professional (OSCP), Nov 2023
Certified Red Team Operator (CRTO), Jun 2023 · Mobile Application Penetration Tester (eMAPT), INE, Feb 2024

AWARDS

Presidential Scholarship, Government of Azerbaijan, for ranking 4th nationwide on the university entrance exam Sep 2019
1st Place, Recyclaton 2022 Hackathon, Coca-Cola Foundation and ASAN Services, with 6,000 AZN seed funding Dec 2022
Golden Medal for Excellency, Ministry of Science and Education of the Republic of Azerbaijan Jun 2019
Bronze Medal, Republican Mathematics Olympiad, Ministry of Science and Education of the Republic of Azerbaijan May 2019

SKILLS

Programming Languages: Python, C, C++, Go, Bash, JavaScript, Java, PHP, SQL
Cybersecurity: Web/Mobile/Network Penetration Testing, Secure Code Review, Binary Exploitation, LLM Red Teaming
Analysis and Exploitation Tools: Burp Suite, ZAP, Metasploit, Nmap, Wireshark, MobSF, Frida, Ghidra, IDA Pro, GDB/pwndbg
Systems and kernel: eBPF, Linux kernel modules, Active Directory tooling (BloodHound, Impacket, Rubeus, Mimikatz)
Cloud and DevOps: AWS (EC2, VPC, RDS, S3, Lambda), GCP, Azure**,** GitHub Actions, Docker, Kubernetes